

# DataCORE

Scientific Storytelling: From Literature Review to Results Language

# Afternoon: Scientific Storytelling

1. The Setup — Your Literature Review as a Story (1:00–1:45)
2. Practice — Your Two Slides, Out Loud (1:45–2:45)
3. The Payoff — Results Language (2:45–3:00)

*This morning you learned the **rules** of academic writing. This afternoon you learn the **craft**: turning your literature review into two slides an audience can follow — and your findings into one sentence they can act on.*

1:00 – 1:45 · The Setup

# Why storytelling? Because nobody remembers a list

Every good research talk is a **story** with the same beats:

1. **Setting** — here's the world and what we know about it (*background*)
2. **Tension** — but here's what we **don't** know (*the gap*)
3. **Quest** — so we asked this question (*research question*)
4. **Resolution** — and here's what we found (*results*)

Today we practice the two moments where storytelling matters most: the **setup** (literature → gap → question) and the **payoff** (results).

 Your audience won't remember your fourth citation. They **will** remember the tension: "*we know A and B — but nobody knows C, so we asked...*"

## The funnel: a chapter becomes two slides

Your literature review is **pages** long. Your presentation gets **two slides**:

### Slide 1 — Background

Key **definitions** + **2–3 cited findings**. What the audience must know to care about your question.

### Slide 2 — Gap + Research Question

What's **missing** from what we know — and the **question** that follows from it.

⚡ This is not "shrinking" your literature review — it's **distilling** it. You keep only what sets up the question. Everything else stays in the written report.

# Anatomy of the Background slide

## 1. Key definitions (1–2 terms)

- Only terms your audience **needs** — not a glossary.
- One line each, in your own words.

## 2. What we know (2–3 cited findings)

- Organized by **theme**, not by author.
- One finding per line — **no paragraphs**.
- Each ends with a citation: **(Author, Year)**.

 **The #1 mistake:** pasting sentences from your written review onto the slide. A slide is a **billboard**, not a page. If it takes more than 5 seconds to read a line, cut it.

# Worked example – the wildfire PBL case

From this morning's climate case: *Does temperature rise affect wildfire frequency?*

## Background: Temperature & Wildfires

**Key terms:** *Acres burned* = total area destroyed by wildfire per year · *Fire season* = months when large fires occur

### What we know:

- Global average temperature has risen about 1.1°C since pre-industrial times (IPCC, 2021)
- Fire seasons in the western U.S. have grown longer and more severe since the 1980s (Westerling, 2016)
- Drought and forest management also shape fire risk — temperature is not the only factor (Abatzoglou & Williams, 2016)

✓ Two terms defined. Three findings, one line each, each cited. A stranger can read this slide in **20 seconds** and be ready for your question.

# Anatomy of the Gap + Research Question slide

## 1. The gap (1–2 sentences)

- What do the studies you cited **not** answer?
- Point back at your Background slide: the gap should feel **inevitable** after it.

## 2. The research question (verbatim)

- Word-for-word the question your team wrote
  - don't paraphrase it live.
- Add your **hypothesis** if you have one.

 The magic sentence connecting the two slides:

*"Because we don't yet know **[gap]**, we ask: **[research question]**."*

If that sentence doesn't flow, one of the two slides needs work.



## Worked example – Gap + Research Question

### The Gap & Our Question

**What's missing:** Most studies track how **long** fire seasons have become. Fewer connect temperature **directly** to how much land actually burns each year over recent decades.

**Our research question:** *Is average summer temperature associated with annual acres burned in the United States, 1990–2020?*

**Our hypothesis:** Years with hotter summers will show more acres burned.

## Worked example – Gap + Research Question

### The Gap & Our Question

**What's missing:** Most studies track how **long** fire seasons have become. Fewer connect temperature **directly** to how much land actually burns each year over recent decades.

**Our research question:** *Is average summer temperature associated with annual acres burned in the United States, 1990–2020?*

**Our hypothesis:** Years with hotter summers will show more acres burned.

Say the magic sentence out loud: *"Because we don't yet know whether temperature tracks **acres burned**, we ask: is average summer temperature associated with annual acres burned?"* → It flows — the slides are doing their job.

## ❌ Four ways these slides go wrong

### **The wall of text**

Full sentences from the written review. Audience reads instead of listens.

### **The name-drop**

"Smith (2020) found... Jones (2021) found..." — organized by author, no theme, no point.

### **The orphan question**

A research question that doesn't follow from the gap — the audience wonders *"why THIS question?"*

### **The missing citation**

A confident claim with no (Author, Year). Instantly costs you credibility.

### ✅ **The fix is the same for all four:**

One line per idea. Themes, not authors. Gap points at the question. Every claim cited.

1:45 – 2:45 · Team Task

## Step 1 – Convert your literature review – ~25 min

With your team, convert your literature review into the **two slides**:

1. **Background slide** – 1–2 key definitions + **2–3 cited findings**, one line each, by theme.
2. **Gap + Research Question slide** – the gap in 1–2 sentences, then your RQ **verbatim** (+ hypothesis).
3. Test with the **magic sentence**: *"Because we don't yet know \_\_\_, we ask \_\_\_."*

### ✓ Quality bar:

- Could a stranger read each slide in **20 seconds**?
- Is every claim **cited** (Author, Year)?
- Does the question feel **inevitable** after the gap?

25:00

## Step 2 – Present to a partner – 10 min

Pair up with someone from **another team**. Each of you:

1. Present your **two slides** in  $\leq 2$  minutes – *speak to your partner, don't read the slides.*
2. Partner gives **two pieces of feedback**:
  - ⚡ *"The clearest moment was..."*
  - 💡 *"I got lost when..."*
3. Swap roles.

 **Listeners:** you are the audience test. If you can't repeat back the gap and the question after 2 minutes, say so – that's the most useful feedback your partner will get today.

10:00

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 Back with your team: apply the feedback to the slides **now**, while it's fresh. These two slides go straight into your final presentation.

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10:00

2:45 – 3:00 · Scientific Storytelling



## Two registers – you need **both**

### **The number** (for credibility)

*"The regression coefficient on experience was 0.34 ( $p < 0.05$ )."*

Other researchers can **check** this.

### **The plain language** (for the audience)

*"Our analysis found that workers earned about 34 cents more per hour for each additional year of experience."*

A policymaker can **act** on this.

 Strong results writing puts the **plain sentence first**, then backs it with the numbers in parentheses. Never make the reader decode statistics to learn what you found.

 You'll learn to **produce** numbers like these in Sessions 8–9. Today you learn how to **say** them.

## Translation practice — say it both ways

Each statistical sentence below is **true but unusable** on its own. Translate it out loud:

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→ *"Our analysis found that men in our sample earned about **\\$.28 more per hour** than women — a gap larger than we'd expect from chance alone ( $p = 0.02$ )."*

2. *" $r = 0.62$  between summer temperature and acres burned."*

## Translation practice — say it both ways

Each statistical sentence below is **true but unusable** on its own. Translate it out loud:

1. *"Mean difference = \ \$2.28,  $p = 0.02$ ."*

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2. *" $r = 0.62$  between summer temperature and acres burned."*

→ *"Our analysis found that **hotter summers went hand in hand with more land burned** — a fairly strong association ( $r = 0.62$ ). Association, not proof of cause."*

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2. *" $r = 0.62$  between summer temperature and acres burned."*

→ *"Our analysis found that **hotter summers went hand in hand with more land burned** — a fairly strong association ( $r = 0.62$ ). Association, not proof of cause."*

 Notice the pattern: **plain claim first** (direction + size in real units), **statistics in parentheses, caveat if needed.**

## Draft your team's headline finding – now

Write **one results sentence** for your team's main finding, in this shape:

*"Our analysis found that **[plain-English finding]** (**[stat]** = \_\_)."*

Then read it to a partner who is **not** on your team. Did they get it?

*No statistics yet? Use your strongest **descriptive** finding – a difference, a trend, a pattern – and put its number in the parentheses.*

### ✓ Checklist:

- Plain-English claim **first**?
- Direction clear (more / less / higher)?
- The number **with units**?
- Statistics in parentheses?
- No causal verb unless you can defend it?

10:00

